DAX MEASURES DOCUMENTATION

Sample Superstore Dashboard — Week 3

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Overview

This document explains the seven DAX measures created for the Week 3 Sample Superstore dashboard, in addition to the base measures (Total Sales, Total Profit) carried over from Week 2. All measures reference the table 'Sample - Superstore'. Each entry below includes the formula, its purpose, a plain-language explanation, and the value it currently returns on the full dataset. One supporting calculated column, OrderYear (= YEAR('Sample - Superstore'[Order Date])), was also added to support the Profit Growth % measure.

The Measures

1. Total Customers

Formula:

```
Total Customers = DISTINCTCOUNT('Sample - Superstore'[Customer ID])
```

Purpose: Counts the number of unique customers in the dataset, rather than the number of orders or line items.

In plain terms: This counts each customer once, no matter how many times they ordered. It answers 'how many different people has the business sold to?' rather than 'how many purchases were made?'

Result in this dashboard: 644 unique customers

2. Total Orders

Formula:

```
Total Orders = DISTINCTCOUNT('Sample - Superstore'[Order ID])
```

Purpose: Counts the number of distinct orders, since a single order can contain multiple product line items.

In plain terms: The raw data has one row per product per order, so counting rows would overcount orders. This measure counts each Order ID only once.

Result in this dashboard: 1,334 orders

3. Average Profit per Order

Formula:

```
Average Profit per Order = DIVIDE(SUM('Sample - Superstore'[Profit]), [Total Orders], 0)
```

Purpose: Shows the average profit earned per order, giving an order-level view of profitability rather than just a company-wide total.

In plain terms: Total profit is divided by the number of orders. The DIVIDE function is used instead of a plain '/' so that the measure safely returns 0 instead of an error if there are ever zero orders in a filtered view.

Result in this dashboard: Approximately $18.71 profit per order

4. Profit Growth %

Formula:

```
Profit Growth % =
VAR CurrentYear = MAX('Sample - Superstore'[OrderYear])
VAR CurrentYearProfit = CALCULATE(SUM('Sample - Superstore'[Profit]), 'Sample -
```

```
Superstore'[OrderYear] = CurrentYear)
VAR PriorYearProfit = CALCULATE(SUM('Sample - Superstore'[Profit]), 'Sample -
Superstore'[OrderYear] = CurrentYear - 1)
RETURN
DIVIDE(CurrentYearProfit - PriorYearProfit, PriorYearProfit, 0) * 100
```

Purpose: Compares profit in the most recent year to profit in the year before it, expressed as a percentage change.

In plain terms: A calculated column called OrderYear (the year extracted from Order Date) is used instead of the DAX DATEADD function, because DATEADD requires a fully continuous calendar and breaks when a slicer creates gaps in the dates in view. CurrentYear finds the latest year present in the current filter context, and the measure explicitly isolates and compares profit for that year against the year before it. A positive number means profit grew; a negative number means it fell.

Result in this dashboard: -47.18% (profit fell from 2016 to 2017)

5. Loss-Making Orders

Formula:

```
Loss-Making Orders = CALCULATE(DISTINCTCOUNT('Sample - Superstore'[Order ID]), 'Sample
- Superstore'[Profit] < 0)
```

Purpose: Counts how many distinct orders resulted in a net loss (negative profit), rather than a profit.

In plain terms: CALCULATE temporarily applies an extra filter — only rows where Profit is less than 0 — and then counts the distinct orders within that filtered set. This isolates unprofitable orders specifically.

Result in this dashboard: 243 orders (18.2% of all 1,334 orders)

6. Total Discount

Formula:

```
Total Discount = SUM('Sample - Superstore'[Discount])
```

Purpose: Adds up the discount values across all rows, giving a total sense of how much discounting activity occurred.

In plain terms: This is a straightforward sum of the Discount column. Note: in the source data, Discount is stored as a rate/fraction per line item, so this total is a summary indicator rather than a currency amount.

Result in this dashboard: 204.49 (summed discount rate across all line items)

7. Discount % of Sales

Formula:

```
Discount % of Sales = DIVIDE([Total Discount], SUM('Sample - Superstore'[Sales]), 0) * 100
```

Purpose: Expresses total discounting as a percentage of total sales, to judge how significant discounting is relative to overall revenue.

In plain terms: Total Discount is divided by Total Sales and multiplied by 100 to express it as a percentage. A low result means discounting plays a small role in the business's pricing; a high result would mean heavy reliance on discounts to drive sales.

Result in this dashboard: 0.07% (discounting is minimal relative to total sales)

Notes on Formatting

The Profit Growth % and Discount % of Sales measures already multiply by 100 inside the formula to produce a percentage-style number (e.g., 28.06 meaning 28.06%). These two measures should be displayed using the

'General' or 'Decimal Number' format in Power BI, not the built-in 'Percentage' format — applying Percentage format on top of a value that is already multiplied by 100 would incorrectly multiply it by 100 a second time.

Summary Table

Measure	Result
Total Customers	644
Total Orders	1,334
Average Profit per Order	$18.71
Profit Growth %	-47.18%
Loss-Making Orders	243 (18.2%)
Total Discount	204.49
Discount % of Sales	0.07%